4.2 Exponential Smoothing: Aggregate-Daily

2026-04-08

Goal

Apply exponential smoothing to the aggregate-daily Rossmann sales series and evaluate its forecasting performance using a holdout set.

We compare the ETS model to the observed holdout period and report holdout RMSE.

```r
library(tidyverse)
library(forecast)

daily_data <- readr::read_csv(
  "../../03_wrangling/daily_all_clean.csv",
  show_col_types = FALSE
) |>
  dplyr::mutate(
    Date = as.Date(Date)
  ) |>
  dplyr::arrange(Date)

head(daily_data)
```

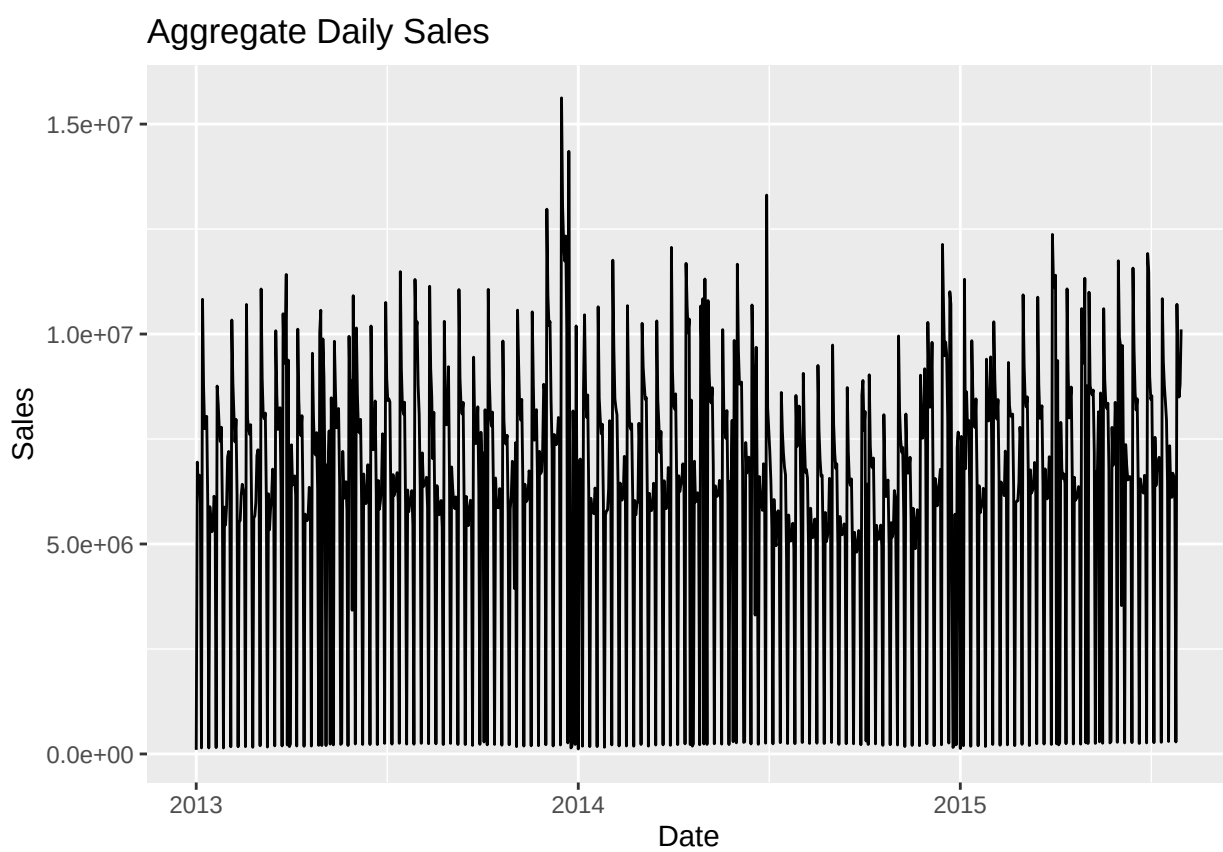
```
## # A tibble: 6 x 8
##   Date         Sales Customers PromoShare SchoolHolidayShare OpenStoreCount
##   <date>       <dbl>     <dbl>      <dbl>              <dbl>          <dbl>
## 1 2013-01-01   97235     19491          0              1                 17
## 2 2013-01-02 6949829    834075          0              1               1111
## 3 2013-01-03 6347820    761040          0              0.933           1109
## 4 2013-01-04 6638954    782137          0              0.933           1108
## 5 2013-01-05 5951593    687472          0              0.100           1107
## 6 2013-01-06  143904     26387          0              0.100             19
## # i 2 more variables: Weekday <chr>, IsWeekend <lgl>
```

```r
holdout_horizon <- 42
n_total <- nrow(daily_data)
n_train <- n_total - holdout_horizon

train_subset <- daily_data |> dplyr::slice(1:n_train)
holdout_subset <- daily_data |> dplyr::slice((n_train + 1):n_total)

holdout_split <- tibble::tibble(
  n_total = n_total,
  n_train = n_train,
  n_holdout = n_total - n_train,
  train_start = min(train_subset$Date),
  train_end = max(train_subset$Date),
  holdout_start = min(holdout_subset$Date),
  holdout_end = max(holdout_subset$Date)
```

```r
)

holdout_split
```

```
## # A tibble: 1 x 7
##   n_total n_train n_holdout train_start train_end  holdout_start holdout_end
##     <int>   <dbl>     <dbl> <date>      <date>     <date>        <date>
## 1     942     900        42 2013-01-01  2015-06-19 2015-06-20    2015-07-31
```

```r
ggplot(daily_data, aes(x = Date, y = Sales)) +
  geom_line() +
  labs(
    title = "Aggregate Daily Sales",
    x = "Date",
    y = "Sales"
  )
```




```r
sales_ts <- ts(train_subset$Sales, frequency = 7)

sales_ts
```

```
## Time Series:
## Start = c(1, 1)
## End = c(129, 4)
## Frequency = 7
##   [1]    97235  6949829  6347820  6638954  5951593   143904 10826848  8749437
##   [9]  7746603  7751485  8039306  6017491   144052  5886788  5512591  5293020
```

```
##  [17]  5386564  6137010  6010192   151600  8761074  8320289  7746563  7439836
##  [25]  7780789  5848770   142848  5878702  5452668  5909779  7068567  7203007
##  [33]  6781741   171565 10333651  8841604  8094803  7430076  7968566  6161019
##  [41]   172564  5517060  5563503  6175147  6422292  6255053  6162142   174467
##  [49] 10703641  7988672  7759768  7611880  7840974  5637945   159057  5615365
##  [57]  5671238  6008503  7108972  7244231  6779350   195229 11072682  8943536
##  [65]  8062941  7981811  8118153  6061383   164917  6194538  5338418  5830737
##  [73]  6098093  6783348  6459058   193072 10078195  8528292  7728892  7897067
##  [81]  8245992  6337911   191036 10482556  9305505  9674698 11417987   213921
##  [89]  9380402   176418   241076  7362049  6406442  6496112  6622062  6171054
##  [97]   189097 10115602  8418477  7698779  7580421  8056347  6024315   182641
## [105]  5711125  5668517  5556697  5726233  6349901  6006147   188406  9542000
## [113]  8047511  7177730  7121467  7652433  6308284   205232 10040182 10565093
## [121]   200972  9884371  8967338  6487719   200337  6884920  6697752  7693180
## [129]   233215  8481406  6658662   215592  9826496  8662237  7767784  7884887
## [137]  8225841  6410461   226808   253936  7204456  6085235  6209485  6483138
## [145]  5806559   201681  9944524  8390312  8885059  3422824 10913876  6907750
## [153]   238144 10144989  8526834  7683188  7646668  7967656  5993792   226250
## [161]  6667460  6201425  5961024  6044207  6887448  6144832   225065 10191372
## [169]  8100286  7243757  7353311  8406280  6189413   224608  6511212  5827831
## [177]  6028195  6372971  7625077  6909121   252185 10750990  8936605  8416942
## [185]  8458325  8384512  5927036   234369  6652999  6157174  6248496  6483182
## [193]  6699439  5987351   253582 11484803  9157994  8269168  8092686  8376476
## [201]  5941848   235211  6285621  5767497  5966755  6065761  6270667  5448670
## [209]   232442 11299276 10064515 10293490  8783485  8085626  6320830   252848
## [217]  7169893  6366382  6414477  6430437  6589299  5881545   243358 11137198
## [225]  9179490  8509505  7029537  8137427  5584570   241793  6391105  5989832
## [233]  5696932  5759988  6035936  5397910   225182 10302422  8397061  7838558
## [241]  7970501  9221577  6458023   250990  6834913  6322579  5985810  5843845
## [249]  6121373  5795003   225500 11058017  8946722  8285894  8096007  8371696
## [257]  5894325   224076  6130906  5785061  5438193  5797033  6042768  5745122
## [265]   204164  9449348  7922441  7390563  7489318  8267396  6245049   228875
## [273]  7660976  6677200  7168893   285518  8196635  6142021   217293 11062400
## [281]  8949093  8089729  7809827  8141538  6203651   227064  6571405  5864402
## [289]  5933839  5861163  6319569  5868927   208367  9833873  7927560  7416700
## [297]  7378657  7589058  5744750   218159  5831815  6145546  6970004  6807674
## [305]  3940492  7410509   175864 10568398  9306571  8224747  7951819  8443830
## [313]  6566741   187852  6423778  6003149  6028677  6140303  6741987  6605960
## [321]   194615 10528615  8895888  7472513  8084449  8199106  6593077   198110
## [329]  7204808  6673704  6725418  7262197  8803269  8614478   220033 12972440
## [337] 10915444 10203870 10296749  8431475  7231141   190125  7607576  7458243
## [345]  7364610  7520198  8012317  7938245   214032 15623548 13053993 12189674
## [353] 11740273 12334496 10165366   263979 14350056  5232808   142585   188597
## [361]  8168568  6726568   226553 10192135  4534864   118108  6980556  7024253
## [369]  6357259   184905  8661727 10456932  8643239  8016642  8548672  6231179
## [377]   175744  6089929  5785842  5779426  5720266  6332625  6213031   175778
## [385] 10648819  8475207  7970289  7625568  7851977  5986034   160153  5689615
## [393]  5790602  5809489  6313068  7935853  6934170   214940 11757740  9651472
## [401]  8456987  8226263  8075318  6387706   190226  6451410  6056843  6046448
## [409]  6180416  7086640  6293698   190158 10675316  8708285  7855918  7738889
## [417]  7862313  6123882   185005  6036002  5697449  5960448  6006428  7874307
## [425]  7088957   225564 10254710  9211960  8802968  8444592  8486290  6075937
## [433]   187110  6204784  5906003  5795093  5898902  6447567  6310072   213125
## [441] 10311703  8426401  7849900  7179418  7681649  6176931   217517  6506590
```

```
## [449]  5909575  5840946  6024721  6822072  6248819   207045 12062933  9114237
## [457]  8314447  8215262  8575005  6560749   223163  6624706  6248167  6410230
## [465]  6524511  6910251  6388099   241475 11683139 10668363 10024850 10357418
## [473]   222365  8428009   186799   240988  6971756  5997676  6039156  6219955
## [481]  5968493   219630 10667908  9457130 10841040   238077 11310699  6850332
## [489]   227506 10797063  9073871  8451824  8356504  8722646  6828031   242817
## [497]  6383523  6128627  6141038  6252912  6515173  6052688   234786 10104320
## [505]  8264365  7727052  7517793  8177009  6130815   224097  6489410  6286135
## [513]  7944392   292254  9846892  6880606   264670 11662688  9598519  8807342
## [521]  8812312  8857173  6731210   275989   288181  7416071  6688764  6939204
## [529]  7068339  6574173   238273 10692037  9295150  9059505  3307766  9683407
## [537]  6065789   230668  6604967  6040970  5977803  5802491  6910940  6586535
## [545]   257513 13308602  8240050  7708595  7204662  6518718  5377184   241423
## [553]  6059587  5119383  4959489  5729302  5791635  5329624   268130  8607714
## [561]  7684994  7177840  6837176  6647315  4689269   248092  5690785  5274595
## [569]  5068787  5179026  5493418  4887412   244638  8536879  7323566  7468663
## [577]  8284756  7544719  5300445   273058  9062750  7530345  6709611  6772684
## [585]  6571862  4938926   249646  5850906  5281785  5149918  5395023  5595129
## [593]  5021534   255808  9250961  7698449  7022798  6616202  6638319  4958030
## [601]   248837  5750763  5056366  5167298  5355611  6562758  5643050   274884
## [609]  9737203  7854744  7170145  6799830  6908744  5010762   228115  5655302
## [617]  5259862  5220667  5288519  5479788  5134569   243543  8724338  7191134
## [625]  6513743  6397477  6550530  4915671   242302  5277645  4977122  4805992
## [633]  4969583  5320736  4975386   220642  8548286  8893788  7702334  8155792
## [641]   307988  6426083   225796  9028634  7332784  6869767  6816808  7047404
## [649]  5190193   240423  5445980  5213744  5110401  5179446  5449686  4945733
## [657]   204991  8078830  6677374  6129832  6463607  6523289  4894704   219359
## [665]  5508341  5163755  5285015  6271893  6106412  3522395   213741  9953793
## [673]  8181287  7493001  7184058  7298281  5629423   177774  8095306  7083145
## [681]  6686356  6750180  7064885  5576429   203453  5849735  5367537  4888765
## [689]  5575878  5816514  5522528   198298  9020315  7868758  7518283  7867510
## [697]  9172722  7305547   242056 10276971  8843822  8258019  8554042  9801856
## [705]  6479401   196469  6560015  6329729  5909802  6156944  6501211  6779226
## [713]   218670 12133381 10621060  9483386  9808212  9518489  7956411   259248
## [721] 11007641 10712124  4612716   155386   205238  5710292   218071  7291426
## [729]  7665743  3895591   130162  7561431  6150311   197193 11303805  6791481
## [737]  8621857  7646645  7945387  6277742   183367  9840664  8538569  7881690
## [745]  7774935  8456148  6525692   195778  6386391  5744162  5882045  5933913
## [753]  6326166  5811369   179028  9401803  8392342  7930037  8024066  9454954
## [761]  7354597   225556 10291172  8904680  8136074  7973916  8439787  6594669
## [769]   208209  6477197  6345089  6393495  6151076  7212357  6701361   217257
## [777]  9321996  8299474  8058390  8031173  8094451  6290922   199419  5984980
## [785]  6021986  6028364  6471718  7778945  7421745   228942 10933146  9234921
## [793]  8421680  8263354  8504508  6565020   201341  6765029  6211021  6272025
## [801]  6376255  6942094  6625997   244505 10877344  8792886  7993269  8073881
## [809]  8291098  6398094   235852  6788834  6226359  6073784  6362734  7078009
## [817]  6829093   226242 12371453 11222913 11102030 11400447   242473  9366558
## [825]   217211   272921  7896155  6790649  6560475  6669239  6445499   244956
## [833] 11075002  9166055  8002511  8396675  8738327  6569186   237342  6588863
## [841]  6039464  6082195  6153519  6367560  6305451   234093 10605500  9302980
## [849]  9517703 11327847   264231  8764984   249207 10995906  8778928  8633577
## [857]  8550616  8665512  7157061   251720  6732629  6686277  8147927   289247
## [865]  8596524  6497847   255680 10603240  8925994  8355497  8228411  8352281
## [873]  6773815   261385   299473  7776413  6876443  6974793  8372820  7546769
```

```
## [881]   278690 11743615  9925193  9731791  3534231  9724893  6478518   262497
## [889]  7366781  6865990  6524741  6553081  6606431  6510261   269533 11570100
## [897]  9636747  8618478  8191358  8458252
```

```r
ets_model <- ets(sales_ts)

ets_model
```

```
## ETS(M,N,M)
##
## Call:
## ets(y = sales_ts)
##
##   Smoothing parameters:
##     alpha = 0.1284
##     gamma = 1e-04
##
##   Initial states:
##     l = 5820592.519
##     s = 1.4163 0.0338 0.9776 1.1732 1.0903 1.1142
##            1.1946
##
##   sigma:  0.2468
##
##      AIC     AICc      BIC
## 31126.44 31126.68 31174.46
```

```r
summary(ets_model)
```

```
## ETS(M,N,M)
##
## Call:
## ets(y = sales_ts)
##
##   Smoothing parameters:
##     alpha = 0.1284
##     gamma = 1e-04
##
##   Initial states:
##     l = 5820592.519
##     s = 1.4163 0.0338 0.9776 1.1732 1.0903 1.1142
##            1.1946
##
##   sigma:  0.2468
##
##      AIC     AICc      BIC
## 31126.44 31126.68 31174.46
##
## Training set error measures:
##                     ME    RMSE     MAE       MPE     MAPE     MASE      ACF1
## Training set -36269.67 1899126 1186058 -101.6393 116.2889 0.567395 0.1705745
```

```r
ets_holdout_fc <- forecast(ets_model, h = holdout_horizon)

holdout_predictions <- tibble::tibble(
  Date = holdout_subset$Date,
```

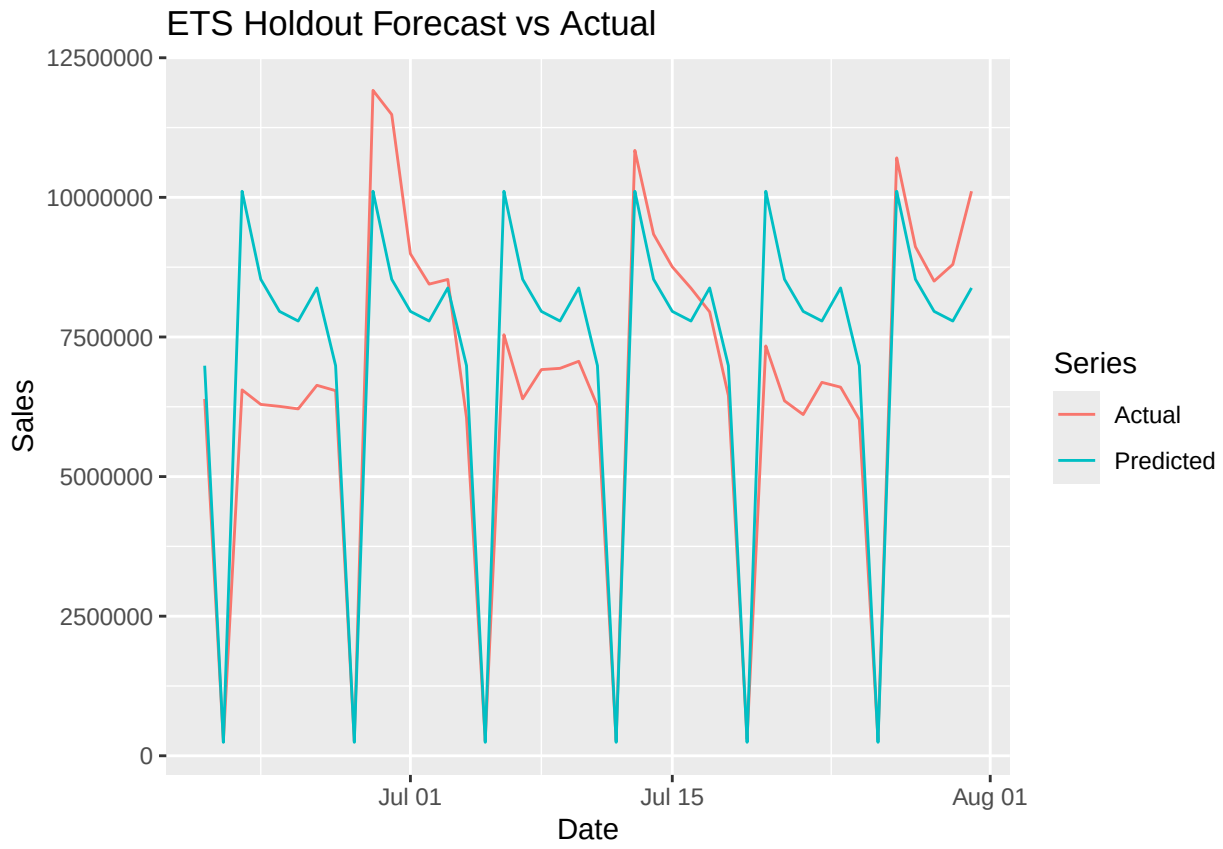
```r
  actual = holdout_subset$Sales,
  predicted = as.numeric(ets_holdout_fc$mean)
)

head(holdout_predictions)
```

```
## # A tibble: 6 x 3
##   Date        actual predicted
##   <date>       <dbl>     <dbl>
## 1 2015-06-20 6390661  6985649.
## 2 2015-06-21  249849   241550.
## 3 2015-06-22 6552538 10109008.
## 4 2015-06-23 6292372  8533010.
## 5 2015-06-24 6257424  7958452.
## 6 2015-06-25 6211764  7785231.
```

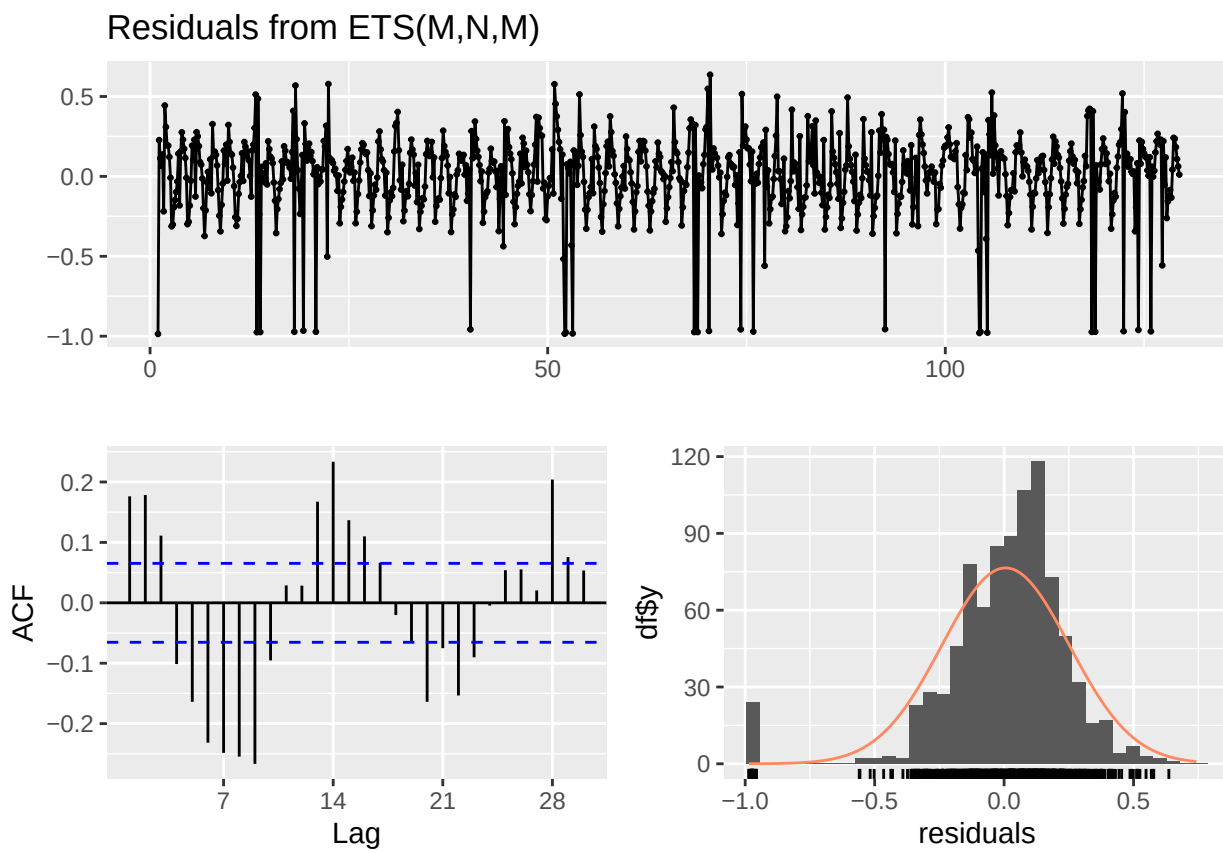
```r
rmse_vec <- function(actual, pred) {
  sqrt(mean((actual - pred)^2, na.rm = TRUE))
}

ets_holdout_rmse <- rmse_vec(
  actual = holdout_predictions$actual,
  pred = holdout_predictions$predicted
)

holdout_rmse_summary <- tibble::tibble(
  model = "ETS",
  holdout_rmse = ets_holdout_rmse,
  n_holdout = nrow(holdout_predictions)
)

holdout_rmse_summary
```

```
## # A tibble: 1 x 3
##   model holdout_rmse n_holdout
##   <chr>        <dbl>     <int>
## 1 ETS       1422687.        42
```

```r
ggplot(holdout_predictions, aes(x = Date)) +
  geom_line(aes(y = actual, color = "Actual")) +
  geom_line(aes(y = predicted, color = "Predicted")) +
  labs(
    title = "ETS Holdout Forecast vs Actual",
    x = "Date",
    y = "Sales",
    color = "Series"
  )
```




```
checkresiduals(ets_model)
```




```
##
##  Ljung-Box test
##
## data:  Residuals from ETS(M,N,M)
## Q* = 415.64, df = 14, p-value < 2.2e-16
##
## Model df: 0.   Total lags used: 14
```

```r
dir.create("results", showWarnings = FALSE, recursive = TRUE)

readr::write_csv(holdout_split, "results/ets_holdout_split.csv")
readr::write_csv(holdout_predictions, "results/ets_holdout_predictions.csv")
readr::write_csv(holdout_rmse_summary, "results/ets_holdout_rmse_summary.csv")
```